

ภาคผนวก ข Component ของ EzVidCap

EzVidCap.ocx

Properties, Methods and Events

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Last Modified: 01/04/98

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Microsoft, Inc copyrights VFWDK and VFW API for Windows.

Properties

AbortLeftMouse Boolean

Set to *True* to allow left mouse button to cancel video capture session

Default value: *True*

AbortRightMouse Boolean

Set to *True* to allow right mouse button to cancel video capture session

Default value: *True*

AutoSize Boolean

Set to *True* to force control to size itself to the video format size

Default value: *True*

BackColor OLE_COLOR

Sets the backcolor of the control

Default value: *&H80000005* (Windows default)

BorderStyle VBCAPBORDERSTYLES

Sets the border style of the control

Default value: *SingleLine* (1)

CancelKey CANCEL_KEY

Sets a keyboard hotkey which will allow user to cancel a capture session from the keyboard

Default value: *KEY_NONE* (0)

CaptureAudio Boolean

Set to *True* to capture audio along with video

Default value: *False*

CaptureFile String

Allows you to set the location and name of the video capture cache file.

*Note** setting this property does not create a file or allocate space. The capture file can be pre-allocated for better performance using the *AllocCapFile* method. Otherwise this file is created, if necessary, when the capture session begins

Default value: *"C:\Capture.avi"*

CaptureRate Long

Set this property to a value between 1 and 100 to request a video capture frame rate from the capture device.

Default value: *15*

Capturing Boolean (Read only)

This property indicates whether a capture session is in progress (*True*)

Default value (when not capturing video): *False*

CaptureViaBackgroundThread Boolean

If this property is *True*, the capture window spawns a separate background thread to perform capture. For example, when you call the *CaptureVideo* function it will return immediately and your code will continue to the next line with no pause.

*Note** Applications that set this flag must handle potential reentry issues because the capture will occur in a separate thread and the application will continue running in the current thread.

Default value: *False*

CenterVideo Boolean

Set to *True* to center the video preview window in the control. This property is useful when *AutoSize* is set to *False*. Otherwise, it has no effect.

Default value: *True*

Container Object

Standard property

DragIcon StdPicture

Standard property

DragMode Integer

Standard property

DriverIndex DRIVERINDEXES

Set this to the VFW driver index of the capture device you want to connect to. The NumCapDevs method should be used to determine the number of installed devices on a system. The GetDriverName and GetDriverVersion methods should be used to return info about devices at each index before connecting.

*Note** the control will automatically attempt to connect to the VFW default index (0) even if this property is not set.

Default value: *VFW_DEFAULT* (0)

FrameEventEnabled Boolean

Set to *True* to tell the control to call the *FrameCallback* event before displaying preview frames. This allows an application to directly access each video frame if desired. The *FrameCallback* event is not raised during streaming video capture.

Default value: *False*

HasAudio Boolean (Read Only)

A value of *True* indicates that there is a WaveIn audio capture device on this system

Default value: *<depends on installed hardware>*

HasDlgDisplay Boolean (Read Only)

A value of *True* indicates that the currently connected VFW capture device will display a standard Display Dialog Box to the user when you call the *ShowDlgVideoDisplay* method. The user can then access and change device specific parameters for this capture device.

Default value: *<depends on installed hardware>*

HasDlgFormat Boolean (Read Only)

A value of *True* indicates that the currently connected VFW capture device will display a standard Format Dialog Box to the user when you call the *ShowDlgVideoFormat* method. The user can then access and change device specific parameters for this capture device.

Default value: *<depends on installed hardware>*

HasDlgSource Boolean (Read Only)

A value of *True* indicates that the currently connected VFW capture device will display a standard Source Dialog Box to the user when you call the *ShowDlgVideoSource* method. The user can then access and change device specific parameters for this capture device.

Default value: *<depends on installed hardware>*

HasOverlay Boolean (Read Only)

A value of *True* indicates that the currently connected VFW capture device supports hardware overlay (allows direct access to the graphics display for improved display speed / lower CPU usage).

Default value: *<depends on installed hardware>*

Height Single

Returns or sets the height of the control

*Note** this property cannot be changed when *AutoSize* is set to *True*.

Default value: *<depends on video format size>*

HelpContextID Long

Standard property

hWnd Long (Read Only)

This property is the hWnd of the control's internal CapCaptureWnd. This value may be used with the functions and macros in the Win32 VFW API in place of a CapWnd hWnd.

Default value: *0* until control is initialized, then *hwnd of internal CapWnd*

Index Integer (Read Only)

Standard property

IndexSize Long

Maximum number of index entries in an AVI file. Values range from 1800 to 324,000. If set to 0, a default value of 34,952 (32K frames plus a proportional number of audio buffers) is used.

*Note**Each video frame or buffer of waveform-audio data uses one index entry. The value of this entry establishes a limit for the number of frames or audio buffers that can be captured.

Default value: *27000* ($30<\text{fps}> * 60<\text{sec}> * 15<\text{min}>$)

Left Single

Standard property

MakeUserConfirmCapture Boolean

If this member is *True*, VFW displays a dialog box prompting the user to initiate capture.

Default value: *True*

Name String

Standard property

Object Object (Read Only)

Standard property

Overlay Boolean

Setting this property to *True* causes VFW to use the hardware overlay features in the installed capture hardware when displaying video in the capture window. The *HasOverlay* property should be checked prior to setting this property to confirm that the hardware actually supports overlay mode.

Default value: *False*

Parent Form (Read Only)

Standard property

PercentDropForError Long

Maximum allowable percentage of dropped frames during capture. Values range from 0 to 100. If the percentage of dropped frames exceeds this value the *Error* event will be raised.

Default value: *10*

Preview Boolean

Enables or disables preview mode.

*Note** Enabling preview mode automatically disables overlay mode.

Default value: *True*

PreviewRate Long

Sets/Returns the frame display rate in preview mode using fps. Valid rates are 1 - 100

*Note** The preview mode uses substantial CPU resources. Applications can disable preview or lower the preview rate when another application has the focus. During streaming video capture, the previewing task is lower priority than writing frames to disk, and preview frames are displayed only if no other buffers are available for writing.

Default value: *15*

StreamMaster CAPSTREAMMASTER

Indicates whether the audio stream controls the clock when writing an AVI file. If this member is set to `AVSTREAMMASTER_AUDIO`, the audio stream is considered the master stream and the video stream duration is forced to match the audio duration. If this member is set to `AVSTREAMMASTER_NONE`, the durations of audio and video streams can differ.

Default value: *AVSTREAMMASTER_AUDIO* (0)

StretchPreview Boolean

Enables or disables scaling of the preview video images. If scaling is enabled, the captured video frame is stretched to the dimensions of the capture window.

*Note** Scaling preview images controls the immediate presentation of captured frames within the capture window. It has no effect on the size of the frames saved to file. Scaling has no effect when using overlay to display video in the frame buffer.

Default value: *False*

TabIndex Integer

Standard property

TabStop Boolean

Standard property

Tag String

Standard property

TimeLimit Long

Time limit for capture, in seconds. This parameter is ignored if `TimeLimitEnabled` is *False*.

Default value: *30*

TimeLimitEnabled Boolean

If this member is *True*, streaming capture stops after the number of seconds in the *TimeLimit* property has elapsed

Default value: *False*

UsePreciseCaptureControls Boolean

Setting this property to *True* gives the user precise control over the capture start and stop times. After setting this property, when the user calls the *CaptureStart* method the control will do all necessary preparations before capture, such as allocating memory buffers. After doing this the control will raise the *PreRollComplete* event and wait for the user to call either the *PreciseCaptureStart* method, to start capturing, or the *PreciseCaptureCancel* method, to cancel the capture session. This method is more complicated than simply calling *CaptureStart*, but it gives the user precise control over the instant that the capture session starts, and it also makes it possible to show a custom user dialog rather than the VFW defined one if desired (see sample application).*Note**

Default value: *False*

VideoBorder VBCAPBORDERSTYLES

Sets the border around the video window to *NONE* or *SINGLE_LINE*.

*Note** This border is separate from the control border

Default value: *SINGLE_LINE*

VideoStreamEventEnabled Boolean

Set to *True* to tell the control to call the *VideoStreamCallback* event before before writing the captured frame to disk. This allows an application to directly access each video frame if desired.

*Note** If this property cannot be changed while video is streaming. It can be disabled after capture ends.

Default value: *False*

Visible Boolean

Standard property

WaveStreamEventEnabled Boolean

Set to *True* to tell the control to call the *WaveStreamCallback* event before before writing the audio buffer to disk. This allows an application to directly access each audio buffer if desired.

*Note** If this property cannot be changed while video is streaming. It can be disabled after capture ends.

Default value: *False*

WhatsThisHelpID Long

Standard property

Width Single

Returns or sets the width of the control

*Note** this property cannot be changed when *AutoSize* is set to *True*.

Default value: *<depends on video format size>*

YieldEventEnabled Boolean

Set to *True* to tell the control to call the *CaptureYield* event during video capture. This event is guaranteed to be called at least once per video frame captured. If this property is set the *CaptureYield* event will be called regardless of the state of the *CaptureViaBackgroundThread* property.

*Note** An application can add a *DoEvents* or process other tasks in this event during video capture.

However extreme care must be taken to handle reentrancy issues which might be caused by accessing the control or related processes during streaming.

Default value: *False*

.Methods

About returns:<none>

Displays an about box with ezVidCap version and copyright information. Takes no parameters.

AllocCapFile returns: Boolean

Creates (preallocates) a capture file of a specified size

Parameters:		
Type	Name	Description
Long	bytes	Number of bytes to allocate
Returns:		
Type	Description	
Boolean	Returns <i>True</i> if successful or <i>False</i> otherwise.	
	Note* If an error occurs, the <i>ErrorMessage</i> event will be raised	

CapSingleFrame returns: Boolean

<takes no parameters>

Retrieves and displays a single frame from the capture driver. After capture, overlay and preview are disabled.

Returns:	
Type	Description
Boolean	Returns <i>True</i> if successful or <i>False</i> otherwise.

CaptureAbort returns: Boolean

<takes no parameters>

Abandons the current capture operation. The capture operation must yield to use this method (set *CaptureViaBackgroundThread* to *True*).

Returns:	
Type	Description
Boolean	Returns <i>True</i> if successful or <i>False</i> otherwise.

CaptureEnd returns: Boolean

<takes no parameters>

Stops the current capture operation. The capture operation must yield to use this method (set *CaptureViaBackgroundThread* to *True*).

Returns:	
Type	Description
Boolean	Returns <i>True</i> if successful or <i>False</i> otherwise.

CaptureVideo returns: Boolean

<takes no parameters>

Initiates streaming video and audio capture to a file.

By default, the capture control does not allow other applications to continue running during capture. To override this, either set the *Yield* property to *True*, or add code to the *Yield Event* in the client application.

During streaming capture, the capture window will issue notifications to your application of specific types of conditions. The *CaptureYield* event will be raised at least once per frame captured if the *YieldEventEnabled* property is set to *True*. The following events may also be raised during capture if the control is either yielding CPU time or running in a background thread:

ErrorMessage

StatusMessage

Returns:	
Type	Description
Boolean	Returns <i>True</i> if successful or <i>False</i> otherwise.

DiskSpace returns: Long

<takes no parameters>

Returns free space on drive which contains capture file in bytes.

Allocated capture file size is not counted as used disk space since it will be overwritten by the control during capture. (for example if there is 10000 bytes free and the capture file size is 5000 bytes, this function will return 15000 bytes).

Returns:	
Type	Description
Long	Amount of free disk space in bytes + size of capture file

Drag returns: <none>

Standard method

EditCopy returns: Boolean

<takes no parameters>

Copies the contents of the video frame buffer and associated palette to the clipboard.

Returns:	
Type	Description
Boolean	Returns <i>True</i> if successful or <i>False</i> otherwise.

GetDriverName returns: String

Returns the driver description string for the currently selected VFW driver. Or if optional Index parameter is passed returns information for device located at the VFW index specified.

Parameters:		
Type	Name	Description
Optional Long	Index	VFW index to return info on (if omitted the current device is queried)
Returns:		
Type	Description	
String	Driver description string for VFW driver. If there is no driver at the VFW index specified, returns "" (Empty string)	

GetDriverVersion returns: String

Returns version string for the currently selected VFW driver. Or if optional Index parameter is passed returns version string for device located at the VFW index specified.

Parameters:		
Type	Name	Description

Optional Long	Index	VFW index to return info on (if omitted the current device is queried)
Returns:		
Type	Description	
String	Version information string for VFW driver. If there is no driver at the VFW index specified, returns "" (Empty string) Note* Some devices do not return version information. In this case the function will return "" (Empty String).	

LoadPalette returns: Boolean

Loads a new palette from a palette file and passes it to a capture driver. Palette files typically use the filename extension .PAL. A capture driver uses a palette when required by the specified digitized image format

Parameters:		
Type	Name	Description
String	palFile	Palette file to load from disk.
Returns:		
Type	Description	
Boolean	Returns <i>True</i> if successful or <i>False</i> otherwise. Note* If an error occurs, the <i>ErrorMessage</i> event will be raised.	

Move returns: <none>

Standard method

NumDevCaps returns: Long

<takes no parameters>

Returns the number of installed VFW capture devices in the system (VFW spec allows for up to 9 devices).

Returns:	
Type	Description
Long	Returns number of video capture devices

PreciseCaptureCancel returns: <none>

<takes no parameters>

Aborts capture session after PreRoll when *UsePreciseCaptureControls* property is set to *True*.

PreciseCaptureEnd returns: <none>

<takes no parameters>

Ends capture session when *UsePreciseCaptureControls* property is set to *True*.

PreciseCaptureStart returns: <none>

<takes no parameters>

Capture session will start at the precise moment this method is called when *UsePreciseCaptureControls* property is set to *True*. Note* This method will have no effect until the *PreRollComplete* event is raised.

SaveAs returns: Boolean

Copies the contents of the capture file to another file.

This message does not change the name or contents of the current capture file.

If the copy operation is unsuccessful due to a disk full error, the destination file is automatically deleted.

Typically, a capture file is preallocated for the largest capture segment anticipated and only a portion of it might be used to capture data. This message copies only the portion of the file containing the capture data.

Parameters:		
Type	Name	Description

String	FileName	Destination filename to to copy data to.
Returns:		
Type	Description	
Boolean	Returns <i>True</i> if successful or <i>False</i> otherwise. Note* If an error occurs, the <i>ErrorMessage</i> event will be raised.	

SaveDIB returns: Boolean

Copies the current frame to a DIB file.

If the capture driver supplies frames in a compressed format, this call attempts to decompress the frame before writing the file..

Parameters:		
Type	Name	Description
String	FileName	Destination filename to to copy data to.
Returns:		
Type	Description	
Boolean	Returns <i>True</i> if successful or <i>False</i> otherwise. Note* If an error occurs, the <i>ErrorMessage</i> event will be raised.	

SavePalette returns: Boolean

saves the current palette to a palette file.

Parameters:		
Type	Name	Description

String	palFile	Destination filename to to copy data to.
Returns:		
Type	Description	
Boolean	Returns <i>True</i> if successful or <i>False</i> otherwise. Note* If an error occurs, the <i>ErrorMessage</i> event will be raised.	

SetFocus returns: <none>

Standard method

ShowDlgAudioFormat returns: Boolean

<takes no parameters>

Displays a dialog which allows the user to select an ACM audio format to use during the capture process.

Returns:	
Type	Description
Boolean	Returns <i>False</i> if no WaveIn audio device is installed on the system (usually means there is no sound card).

ShowDlgCompressionOptions returns: Boolean

<takes no parameters>

Displays a dialog box in which the user can select a compressor to use during the capture process. The list of available compressors can vary with the video format selected in the capture driver's Video Format dialog box.

Software compression is only used with capture drivers that provide frames in the BI_RGB format. This message is most useful in the step capture operation to combine capture and compression in a single

operation. Compressing frames with a software compressor as part of a real-time capture operation is most likely too time-consuming to perform.

Compression does not affect the frames copied to the clipboard.

Returns:	
Type	Description
Boolean	Returns <i>True</i> if successful or <i>False</i> otherwise.

ShowDlgVideoDisplay returns: Boolean

<takes no parameters>

Displays a dialog box in which the user can set or adjust the video output. This dialog box might contain controls that affect the hue, contrast, and brightness of the displayed image, as well as key color alignment.

The controls in this dialog box do not affect digitized video data; they affect only the output or redisplay of the video signal.

The Video Display dialog box is unique for each capture driver. Some capture drivers might not support a Video Display dialog box. Applications can determine if the capture driver supports this method by checking the *HasDlgDisplay* property.

Returns:	
Type	Description
Boolean	Returns <i>True</i> if successful or <i>False</i> otherwise.

ShowDlgVideoFormat returns: Boolean

<takes no parameters>

Displays a dialog box in which the user can select the video format. The Video Format dialog box might be used to select image dimensions, bit depth, and hardware compression options.

The Video Format dialog box is unique for each capture driver. Some capture drivers might not support a Video Format dialog box. Applications can determine if the capture driver supports this method by checking the *HasDlgFormat* property.

Returns:	
Type	Description
Boolean	Returns <i>True</i> if successful or <i>False</i> otherwise.

ShowDlgVideoFormat returns: Boolean

<takes no parameters>

Displays a dialog box in which the user can control the video source. The Video Source dialog box might contain controls that select input sources; alter the hue, contrast, brightness of the image; and modify the video quality before digitizing the images into the frame buffer.

The Video Source dialog box is unique for each capture driver. Some capture drivers might not support a Video Source dialog box. Applications can determine if the capture driver supports this method by checking the *HasDlgSource* property.

Returns:	
Type	Description
Boolean	Returns <i>True</i> if successful or <i>False</i> otherwise.

StreamNoFile returns: Boolean

<takes no parameters>

Initiates streaming video capture without writing data to a file.

This function is useful if you want to access video or audio data in the *VideoStreamCallback* event or *WaveStreamCallback* event without writing data to a file.

By default, the capture control does not allow other applications to continue running during capture. To override this, either set the *Yield* property to *True*, or add code to the *Yield Event* in the client application.

During streaming capture, the capture window will issue notifications to your application of specific types of conditions. The *CaptureYield* event will be raised at least once per frame captured if the *YieldEventEnabled* property is set to *True*. The following events may also be raised during capture if the control is either yielding CPU time or running in a background thread:

ErrorMessage

StatusMessage

Returns:	
Type	Description
Boolean	Returns <i>True</i> if successful or <i>False</i> otherwise.

ZOrder returns: <none>

Standard method

Events

CaptureYield

This event is raised when the capture window yields during streaming capture.

Parameters:	
Type	Description
<none>	
Remarks:	
See documentation for <i>YieldEventEnabled</i> property	

DragDrop

Standard event.

DragOver

Standard event.

ErrorMessage

This event is raised when the capture window yields during streaming capture.

Parameters:		
Type	Name	Description
Long	ErrCode	An IDS_CAP* error code (see VFW.bas by Ray Mercer for VB translations of the Microsoft VFW.H)
String	ErrString	A String describing the ErrorMessage
Remarks:		
The ErrString parameter is defined by the VFW system, therefore the strings will vary depending upon the O.S. language version. Use the ErrCode parameter if you need consistency across international Windows platforms.		

FrameCallback

This event is raised when the capture window captures preview frames.

Parameters:		
Type	Name	Description
Long	lpVHdr	A pointer to a VIDEOHDR type which contains information about the captured frame. (see VFW.bas by Ray Mercer for VB translations of the Microsoft VFW.H)
Remarks:		
This version of the ezVidCap.ocx does not provide guaranteed access to this data before the frame is shown in real time. Development is currently underway for this feature to be implemented in a future version.		

GotFocus

Standard event.

LostFocus

Standard event.

PreRollComplete

If the *UsePreciseCaptureControls* property is set to *True* this event will be raised when the *CaptureVideo()* method is called after all buffers are allocated and preroll activity is complete. After this event is raised, you must either call *PreciseCaptureStart()* or *PreciseCaptureCancel()* to start or cancel a capture session, respectively.

Parameters:		
Type	Name	Description
<none>		
Remarks:		
This event is not raised unless <i>UsePreciseCaptureControls</i> is set to <i>True</i> before <i>CaptureVideo()</i> is called. See the documentation for <i>UsePreciseCaptureControls</i> for more information.		

StatusMessage

This event is raised when VFW sends a status message to the control. Some status messages include both a numeric code and a native VB string, others do not return a description string.

Parameters:		
Type	Name	Description
Long	StatCode	An integer which indicates a status change in the control
String	StatString	A String description of the current status (not all status changes fill this parameter – if there is no description string for the code being sent this parameter will be “” – empty string)
Remarks:		
A list of VFW status messages is included in the file VFW.bas by Ray Mercer. This file is freely downloadable from http://i.am/shrinkwrapvb		

VideoStreamCallback

This event is raised during streaming capture when a video buffer is filled.

Parameters:		
Type	Name	Description
Long	lpVHdr	A pointer to a VIDEOHDR type which contains information about the captured frame. (see VFW.bas by Ray Mercer for VB translations of the Microsoft VFW.H)
Remarks:		
This version of the ezVidCap.ocx does not provide guaranteed access to this data before the frame is written to disk. Development is currently underway for this feature to be implemented in a future version.		

WaveStreamCallback

This event is raised during streaming capture when a new audio buffer becomes available.

Parameters:		
Type	Name	Description
Long	lpWHdr	Pointer to a WAVEHDR structure containing information about the captured audio data
Remarks:		
This version of the ezVidCap.ocx does not provide guaranteed access to this data before the data is written to disk. Development is currently underway for this feature to be implemented in a future version.		